

The path, not the end state: household fleet simulation, estimated behaviour and exploratory backcasting for robust and acceptable low-emission transport

Arshad Akhtar Abbasia · PhD in Logistics, Molde University College · TRANSPLAN research areas 1 to 3 · case studies: Molde and Kristiansund

Working prototype and every number in this proposal: github.com/Astheno-sphere/computational_Logistics

why now

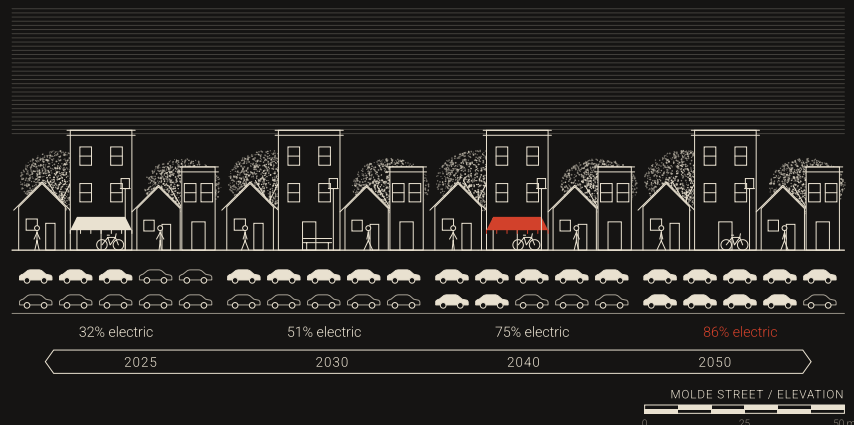
1 · THE ROAD, NOT THE DESTINATION

Norway won the showroom. The fleet and the politics have not caught up.

In 2025, 95.9% of new cars sold in Norway were fully electric¹, yet most cars on the road still burn fuel, and a car stays in the fleet for close to two decades. Fuel tax revenue is eroding, and the distance-based charge meant to replace it has stalled between the state and the municipalities². The National Transport Plan 2025 to 2036 commits to better methods for steering toward the 2050 low-emission society³, and the Ministry's methods assignment admits that today's models suit continued policy, not deep cuts⁴. TRANSPLAN was created to work backwards from the 2050 goals⁵.

TØI's own review names the gap in one line: RTM and NTM6 compute partial equilibria but "say nothing about the path to a new equilibrium"⁶. **Backcasting is a path**, and that path runs through households: the car each one owns, the year it is replaced, the trips it makes and the charges it will accept. My PhD builds the engine that gives TRANSPLAN that path, driven by preferences estimated in a stated-choice survey, and tests whether the plans built on it survive futures nobody can forecast. The households are commuters, but the network they share is the one goods are delivered on, which is where the work meets logistics.

the stock problem, in one street



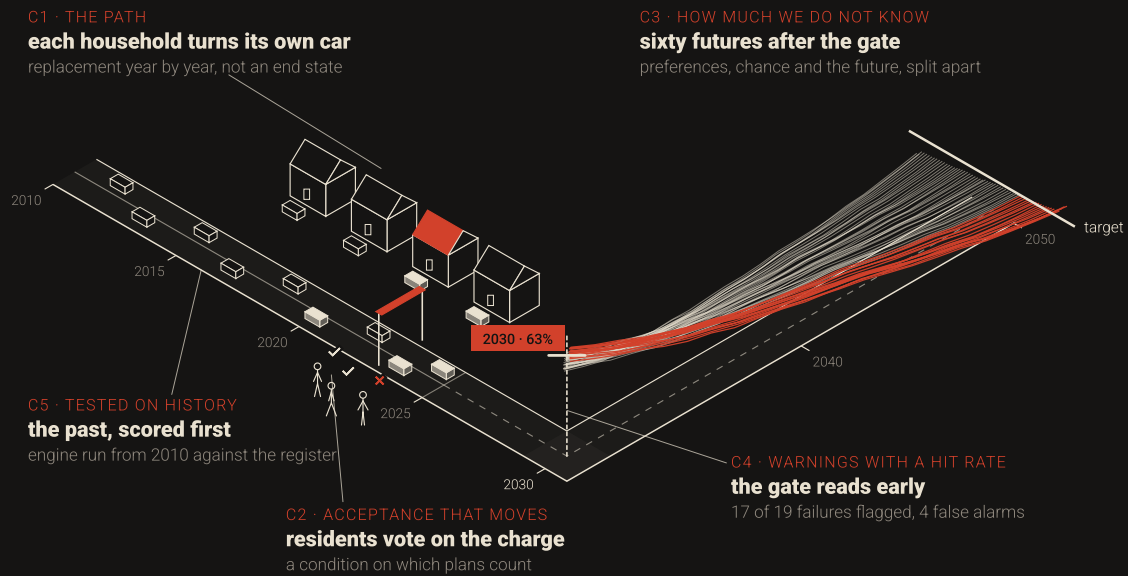
THE STOCK TURNS

one street, four moments. each lit car is electric, in the share the household engine simulates for that year under the most robust package. the street changes only as fast as households replace the cars they already own.

Prototype, median future, package P12; without policy the 2050 share is 74%. The prototype's purchase model is not yet calibrated to Norwegian sales and turns the fleet more slowly than TØI's BIG model; A2 calibrates it to the register. The plate shows the mechanism, not the expected fleet.

2 · FIVE CONTRIBUTIONS

the five contributions, on one road



THE ROAD SPLITS AT 2030

time runs along the road. the engine is first scored on the record (C5). households turn the fleet one car at a time (C1) and vote on the charge they drive under (C2). at the 2030 gate one reading warns early (C4). after the gate the road opens into sixty futures, and a plan counts only if it holds in most of them (C3).

Fan: prototype ensemble, package P12, 60 futures, CO₂ relative to 2025; red futures miss the 2050 target. Synthetic behaviour.

C1 · THE PATH

A household engine that shows how Norway gets from today to 2050, year by year

Agents carry each household's car, its age, its replacement year, its trips and its response to price. Norway's fleet model BIG follows vintages with an estimated choice model, but in aggregate and for fixed scenarios⁷.

nearest: Fridstrøm et al. 2016; Deuten et al. 2020

C2 · ACCEPTANCE THAT MOVES

Acceptance measured in choice data, allowed to shift once a charge is lived, and used as a condition on which plans count

Nationally, 70 to 80% support moderate distance charges with earmarked revenue, least among drivers with few alternatives⁸. Support rose after Stockholm's charge⁹ but moved little in the Oslo field trial⁸, so the shift is tested, not assumed. Le Pira, Marcucci and Gatta gave freight agents acceptance¹⁰; I carry it into passenger policy and into the robust search itself.

nearest: Le Pira et al. 2017; Ciccone et al. 2026

C3 · HOW MUCH WE DO NOT KNOW

2050 uncertainty split into three sources, and a test of whether the winning plan survives each

Error in the estimated preferences, chance inside the simulation and deep uncertainty about the future, separated over 25 years. Petrik and colleagues propagated parameter error through one simulated day¹¹; Tveter, Welde and Odeck quantify appraisal uncertainty for single road projects¹². I separate the sources along a policy path.

nearest: Petrik et al. 2020; Tveter et al. 2025

C4 · WARNINGS WITH A HIT RATE

Pathways stress-tested across thousands of futures, with signposts whose lead time and false-alarm rate are measured

Measured signpost detection appears mainly in water management^{13,14}; transport backcasting remains national and deterministic¹⁵. Each signpost gets years of warning, false alarms and missed failures.

nearest: Raso et al. 2019; Haasnoot et al. 2018

C5 · TESTED ON HISTORY

The engine is checked against Norway's 2010 to 2025 electric transition before it is trusted with 2050

EV forecasts often fail against what happened¹⁶, and on Norwegian fixed links, long-run traffic was in some cases twice the forecast¹⁷. Deuten et al. tested a national system dynamics model on 2010 to 2017¹⁸; I score a household engine at zone level against the register.

nearest: Deuten et al. 2020

where

3 · TWO TOWNS, ONE MODEL

Molde and Kristiansund are **case studies, not a national sample**. They are car-dependent towns of the kind where the national experiment found charges hardest to accept⁸, and tolls, bus service, parking and cycling are decided locally. Their networks concentrate travel: routing every pair of junctions by its fastest path, 17.5% of all routes in Molde use Øvre veg and 24.9% in Kristiansund cross Gomabrua, so one charge or bus lane changes travel for most of the town at once. Molde is the main case, where preferences are estimated and the engine calibrated. Kristiansund is the transfer case: the Molde model is applied without re-estimation, which turns the transferability of preferences from an assumption into a measurement. Inputs are national and outputs are reported against national targets, so the method carries to other regions. The same road graph already solves delivery problems for Molde in the repository.

how

4 · ONE ENGINE, THREE ARTICLES

The engine has four layers: preferences estimated from people, the real road network, households carried through time, and the futures every plan is tested against. Each article builds one layer on the one below and hands a defined product to the next.

A1 · TRANSPLAN RA3 · YEAR 1

Preferences and acceptance

How do people in Molde and Kristiansund trade off time, cost and service, when do they replace their car, and which charging packages would they accept? Does national acceptance hold in car-dependent towns, and do preferences transfer between them?

Design. A stated-choice survey with three blocks: mode choice, timing and type of the next car, and votes on charging packages that vary charge level, exemptions, alternatives offered and use of revenue, with attribute ranges matched to the national experiment⁸. Each respondent completes two of the three blocks, assigned at random, with four tasks in each (about 15 minutes), so every block has about 530 respondents and enough repeated choices to estimate random-parameter spreads. An efficient design is piloted on about 50 respondents, and the sample of about 800 follows from the design's S-estimates for every parameter, random-parameter spreads included, with priors updated after the pilot.

Recruitment. A Norwegian web panel combined with postal invitations to a random sample from the population register, so coverage does not rest on panel members in two small towns; at an assumed 15% response, about 5,500 invitations. Hypothetical bias is reduced by a cheap-talk script, a consequentiality statement and levels pivoted on each respondent's own car and commute. **Models.** Mixed and latent-class logit, willingness to pay by segment, a hybrid choice model in which trust, measured by attitudinal indicators, shapes acceptance, and stated replacement timing estimated jointly with revealed replacement from the register, so its scale is anchored to observed behaviour. **Product.** Preferences with their covariance and an acceptance function the agents carry.

A2 · TRANSPLAN RA2 · YEAR 2

The household fleet engine

When estimated preferences drive households with real cars, how do fleet renewal, mode shift, acceptance and who pays evolve year by year, and how much of the 2050 spread comes from preferences, from chance and from the future?

Engine. Households hold cars of known age and fuel, replace them by the A1 choice model, choose modes on the OpenStreetMap network, and hold opinions on charges whose shift with experience is set from the field evidence^{8,9} and reported with and without it; peer influence is bounded by sensitivity analysis, not claimed as identified. Replacement depends on car age, purchase is calibrated to register extracts at zone level requested from the Norwegian Public Roads Administration in month 1, and preference parameters are drawn from their estimated distribution with repeated runs, so chance and preference error separate. **Validation.** Calibration matches several observed patterns at once (travel survey mode shares, fleet age and fuel from the register, traffic counts)¹⁹, with Kristiansund held out and 2010 to 2025 as a historical test that holds the 2026 preferences fixed and so scores the fleet mechanics, documented with the ODD protocol²⁰. **Scope and links.** Commuting and other regular car travel by residents, reported on that basis and translated to national targets. Emissions are well to wheel with Norwegian grid factors, rebound included, and vehicle and battery production are charged at purchase; welfare is consumer surplus, treated by the Ministry of Finance appraisal guidance including the cost of public funds; zone data come from RTM. The engine is built in year 1 on synthetic preferences, so year 2 swaps in real ones rather than starting from zero. **Product.** A calibrated engine and an ensemble interface.

Robust pathways and warnings

Across thousands of futures, which packages stay effective, acceptable and affordable, when do they fail, and which 2030 reading warns early, with what lead time and false-alarm rate?

Decision problem. Robustness $R(p)$ is the share of futures in which commuting emissions in 2050 fall to τ of the base year; the search returns $p^* = \operatorname{argmax} R(p)$ subject to acceptance $A(p) \geq \frac{1}{2}$ in the median future and public cost $C(p) \leq B$, with B set with the county from its transport budget. I use a satisficing measure because the 2050 goal is itself a threshold to be met, not a quantity to maximise, and because it needs no probabilities over futures nobody can assign. Road traffic emissions are close to their 1990 level²¹, so the Climate Change Act's economy-wide goal of a 90 to 95% cut implies τ of about 5 to 10% of 2025 for commuting; the prototype uses a looser 20%, and the thesis reports 5, 10 and 20%, with regret and stability beside robustness, so any ranking that depends on the metric is shown rather than hidden.

Method. About 10^4 futures anchored on the national strategies¹⁵ and 10^2 packages of the instruments debated in 2026: distance charging, toll rings in which electric cars pay, transit frequency and fares, cycling and fleet renewal. Packages are screened as a Latin hypercube sample and the best region refined by a many-objective search²², with scenario discovery²³ and Sobol indices. One prototype run takes about 0.1 s with 1,000 agents; the full design is costed from measured run times on national computing resources, with a surrogate if one full-scale run passes ten seconds. Signposts are chosen on one set of futures and scored on a held-out set for lead time, false alarms and missed failures¹³, and pathways for how often they must change²⁴. **Freight test.** The search is rerun with urban deliveries on the same network, using the route solver already in the toolkit, so each package is also scored on delivery kilometres and emissions. Delivery demand is held fixed, so the test isolates how a package changes delivery routes, with emissions counted on the same basis as the cars. It tests that the method carries to logistics; the thesis does not depend on a freight demand model. **Product.** Robust, acceptable pathways with measured signposts.

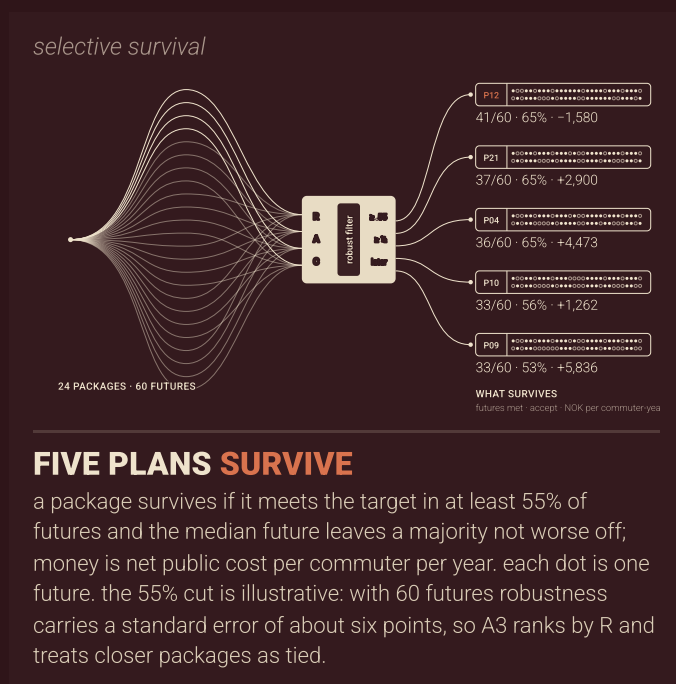
A4 · OPTIONAL

A workshop with county and municipal planners tests whether pathway stories help them agree on the next decision²⁵.

proof

5 · IT ALREADY RUNS

A working prototype runs the full chain on synthetic behaviour and the real networks of both towns: two choice-model estimators with a recovery test, an agent model with fleet turnover, welfare and public money per commuter, and 1,500 runs of 24 packages and a no-policy reference in 60 futures. The behaviour is synthetic, replacement is a constant yearly rate, every package shares one random draw, acceptance is a self-interest proxy and the budget is not yet applied, so the numbers show what each article will produce, not what will happen in Norway. The code is public, 87 automated tests run on every change, and every number here is rebuilt by committed scripts.



Prototype ensemble, 1,500 runs; target: commuting CO₂ in 2050 at most 20% of 2025. Synthetic behaviour.

when

6 · PLAN, RISKS AND ETHICS

	YEAR 1	YEAR 2	YEAR 3
Research	Data applications in month 1; survey design, pilot, fieldwork; A1. Engine built in parallel on synthetic preferences	Real preferences swapped in; calibration, Kristiansund transfer, 2010 to 2025 test; A2	Ensemble, robust search, held-out signposts, freight test; A3; thesis
Training	PhD courses: stated choice, econometrics, simulation	Research stay with a deep-uncertainty group	Optional A4 with planners through TRANSPLAN

Risks. Low response: web panel plus register sample; if fieldwork runs past month 9, A1 is submitted on the pilot and panel sample and the postal wave feeds A2. Delayed data: Statistics Norway commuting tables. Run time: surrogates and national computing. Weak identification: several observed patterns at once and a hold-out town. Results that hinge on the target or metric: reported across τ and metrics. **Ethics.** The survey is assessed by Sikt before fieldwork, with informed consent and pseudonymised storage; code is open under MIT, and data are released with a DOI where consent allows.

what changes

7 · FOR TRANSPLAN, THE REGION AND THE FIELD

For TRANSPLAN, a behavioural experiment for research area 3, the transition path national models lack for research area 2, and measured pathways for the backcasting of research area 1, built to sit alongside NTM6 and RTM and to move from “predict and provide” to “decide and provide”^{26,27}. **For Møre og Romsdal**, pathways with signposts planners can read from traffic counts and the vehicle register. **For the field**, acceptance as a moving condition on robustness, uncertainty split into its sources, and signposts with measured hit rates in transport.

The fleet that will drive in 2035 is being bought now. This project builds the instrument to steer it, and to know early when a plan needs to change.

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